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IOT BASED SMART PLANT MONITORING AND CONTROL SYSTEM

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INTRODUCTION

- This project is an IoT Based Smart Plant Monitoring and Control System.
- Plants need the right amount of water for healthy growth.
- It helps to reduce water wastage and minimizes manual efforts.
- Manual watering can cause overwatering or underwatering.
- As urbanization increases and lifestyles become busier, maintaining healthy plants can be challenging.

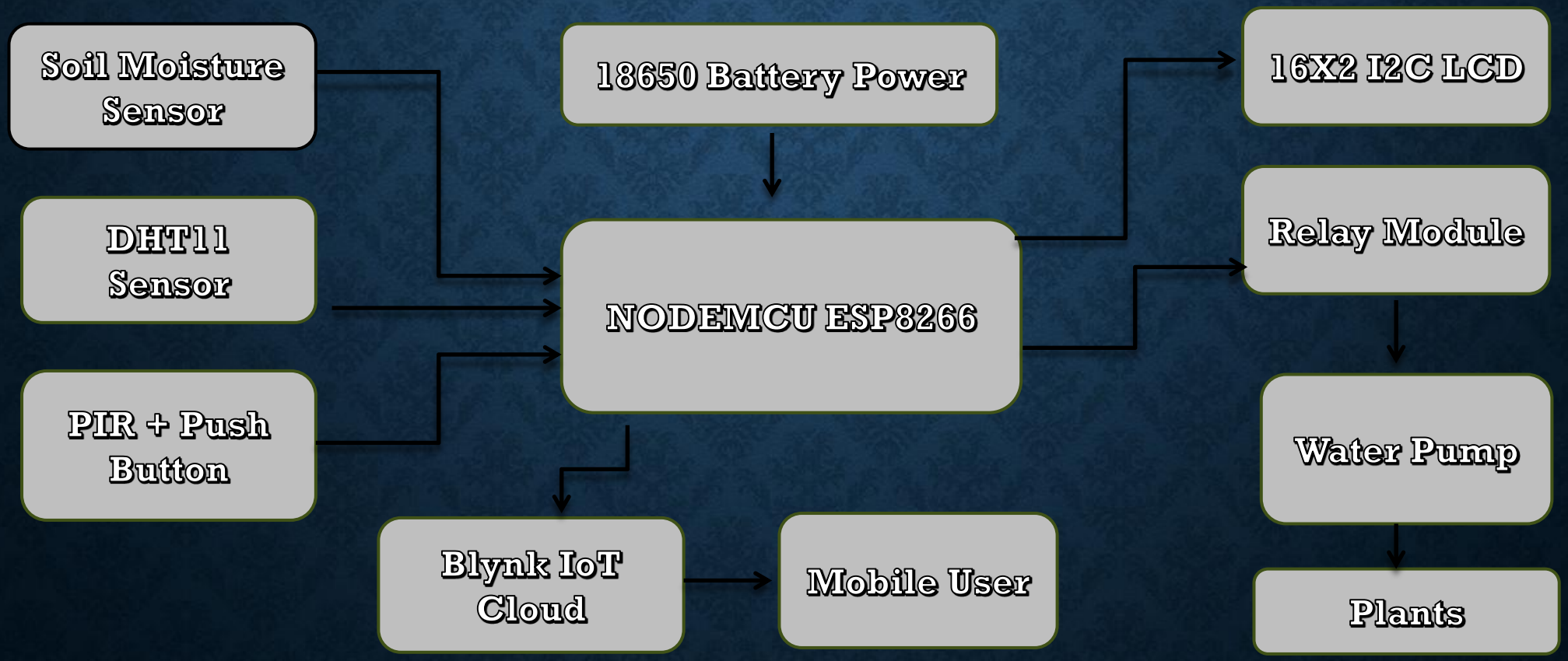
OBJECTIVE

- To Detect soil moisture and supply water automatically.
- To monitor plant conditions using IoT technology.
- To reduce water wastage by smart irrigation.
- To allow remote monitoring and control through mobile app.
- To reduce human effort and improve efficiency.

LITERATURE AND METHODOLOGY

- Traditional irrigation requires manual effort causes water wastage, so IoT based automatic systems are used.
- Sensors like soil moisture and temperature are used to monitor plant conditions.
- NodeMCU processes the sensor data and compares it with a predefined threshold.
- If soil is dry the pump turns ON, and if soil is wet the pump turns OFF automatically.
- The system displays data on LCD and sends real-time updates to the blynk mobile app continuously.

BLOCK DIAGRAM



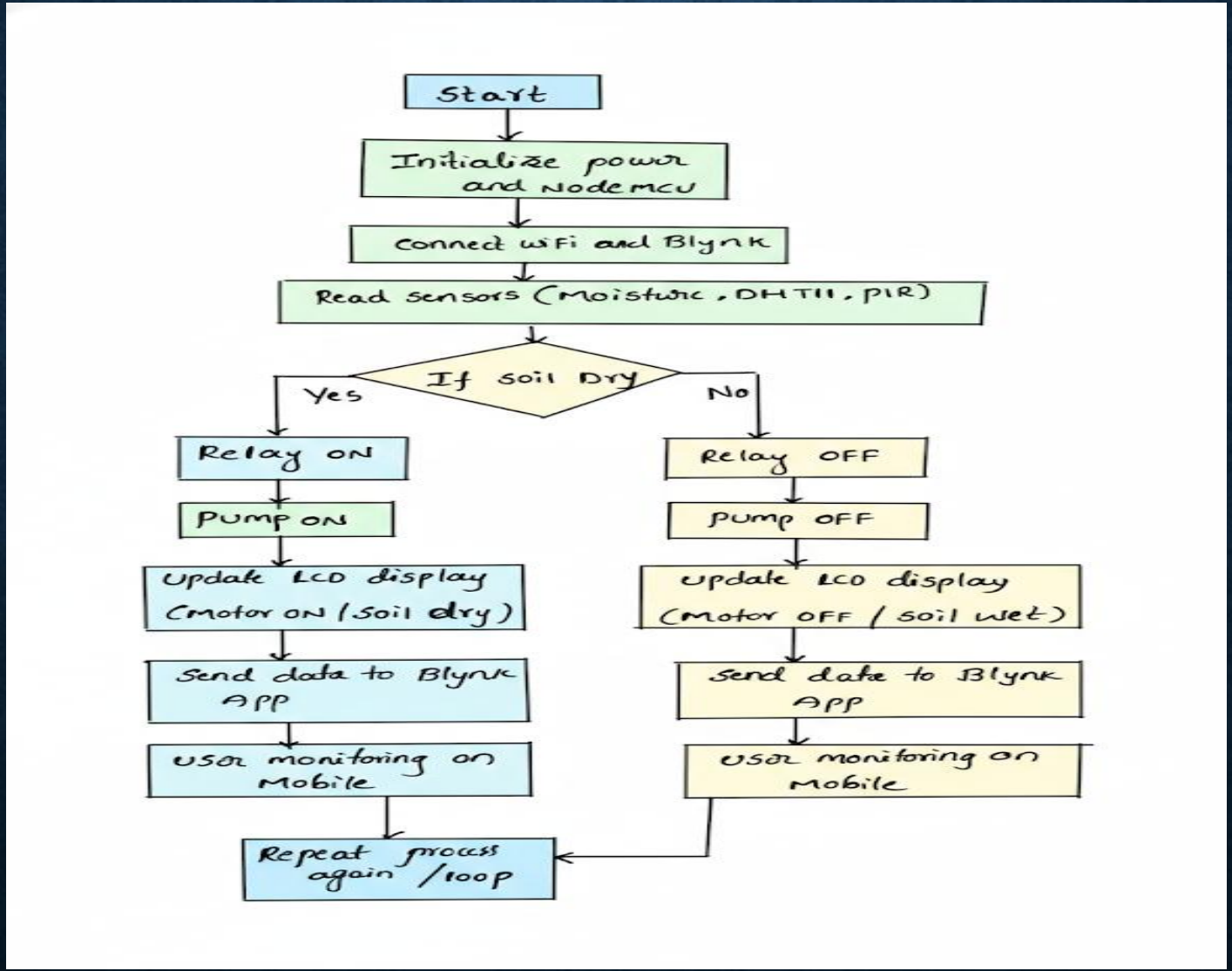
HARDWARE COMPONENTS

- **NODEMCU ESP8266** : Main controller, processes data and connects to WiFi.
- **Soil Moisture Sensor** : Detect soil wet or dry condition.
- **DHT11 Sensor** : Measures temperature and humidity
- **PIR Motion Sensor** : Detects movement near plant.
- **Relay Module** : Controls ON/OFF of water pump.
- **Water Pump** : Supplies water to the plant.
- **16×2 I2C LCD Display** : Shows sensor values and status.
- **18650 Battery** : Power supply for the system.
- **Push Button** : Manual control of pump.
- **Breadboard & Jumper Wires** : Circuit connections.

SOFTWARE

- Arduino IDE – Used to write and upload code to the NODEMCU ESP8266.
- Embedded C / Arduino Programming Language – Used to implement the system logic.
- ESP8266 WIFI Libraries – Used to establish internet connectivity.
- Blynk IoT Mobile Application – Used for remote monitoring and control.

FLOW CHART


Start
Initialize power and Nodemcu
Connect wiFi and Blynk
Read sensors (Moisture, DHT11, PIR)
If soil Dry
Yes
No
Relay ON
Pump ON
Update LCD display (motor ON / soil dry)
Send data to Blynk App
User monitoring on Mobile
Relay OFF
Pump OFF
Update LCD display (motor OFF / soil wet)
Send data to Blynk App
User monitoring on Mobile
Repeat process again / loop

APPLICATIONS

- **Home gardens** : Simplifies the care of indoor and outdoor plants.
- **Agriculture** : Used in large-scale farming to optimize water usage and crop yield.
- **Greenhouses** : Maintains optimal growing conditions for plants.
- **Potted plants** : Ideal for busy individuals who want to maintain plant health without daily attention.
- **Parks and Offices :** Useful for maintaining plants in public places with minimal human effort.

ADVANTAGES

- Time saving
- Water Conservation
- Improved Plant Health.
- Customization
- Low Maintenance.
- Cost Effective.

REFERENCES

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- ESP8266 NodeMCU Datasheet. https://www.espressif.com/sites/default/files/documentation/esp8266-technical_reference_en.pdf
- Blynk IoT official website- https://blynk.io
- Soil Moisture Sensor and DHT11 Sensor Datasheets. https://components101.com/sensors/dht11-temperature-sensor
- IoT Based Smart Irrigation research papers. https://scholar.google.com (Search: *IoT Smart Irrigation System*)

Thank
You